Solvefy — Test Task

AI Engineer I (Associate) · 0–1 year experience

Time to complete: 7 calendar days from the date you receive this. **Submission deadline:** 24 hours before your scheduled interview.

We are not going to tell you how many hours to spend. That decision is yours, and what you do with it is part of what we are looking at. All we ask is that you tell us, in your README, roughly how long you spent. There is no right answer, and a bigger submission is not automatically a better one — deciding how much to build is a real engineering judgement and we would rather see yours than give you ours.

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1. Read this part first

This is not a test of whether you can write code without AI. We expect you to use AI for most of the typing. **We are evaluating how you work** — how you turn a vague problem into a clear one, how you decide what to build, how you direct AI, and how much you understand of what comes back.

Three things follow from that:

You must use the BMAD Method. Use it to build your requirements and your context before you build the software. BMAD — the Breakthrough Method for Agile AI-Driven Development — is an open method, and we have given you the starting points below. How far you get with it in seven days tells us something we cannot learn any other way.

Where to start with BMAD

Official masterclass video	https://www.youtube.com/watch?v=LorEJPrALcg
Official channel	https://www.youtube.com/@BMadCode
Documentation	https://docs.bmad-method.org/
Source repository	https://github.com/bmad-code-org/BMAD-METHOD

Start with the masterclass video — it is the clearest single explanation of the whole workflow. One thing to be aware of: that video is over a year old and the method has moved on since it was recorded, so some of what you see will not match the current documentation or the current release. Work out what has changed and follow the current version. Noticing that for yourself is part of the exercise.

You must submit every AI conversation you used. Complete and unedited, including the parts that went badly. A submission without conversation logs will not be interviewed. We are not looking for perfect prompts — we are looking at how you recover when the AI gets it wrong, which is the most useful thing we can possibly see.

The requirements below are deliberately incomplete. This is what real client requirements look like. Some things are vague, some are missing, and at least one thing contradicts something else. **Do not contact us for clarification — we will not answer.** Make your own decisions, write them down, and be ready to defend them.

In the interview we will ask you why you chose option A over option B. "The requirements didn't say" is not an answer we accept; deciding well with incomplete information is the actual job.

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2. The client

Horizon Survey Services — a land-surveying company in Lahore. About 25 people, of whom roughly 14 are field staff. They own expensive equipment: total stations, GPS rovers, tripods, drones, laptops, field tablets.

You have two pieces of input from them. That is all you get.

2.1 Transcript of a voice note from the owner, Mr Tariq

> "So basically the problem is the equipment. We have a lot of equipment, it is expensive, and I never know where any of it is. The boys come in the morning, they pick up what they need, they go to site. Fine. Then somebody else needs the same thing and nobody knows who has it. Last month we bought a second GPS rover for eleven lakh because we thought ours was lost. It was in somebody's car boot for three weeks.
>
> So I want to know who has what. That's it really. And I want it back on time, people keep things for weeks when they needed it for one day.
>
> Anyone in the team can take anything, we don't have a system, it's all on trust, and mostly that's fine, I don't want to make it like the army. But I want to see it on a screen.
>
> Sometimes they take the whole survey kit, sometimes just one piece. Oh and the office needs to know too, Saima in the office gets the calls asking where things are, she is the one suffering.
>
> One more thing — the boys are not going to use anything complicated. Half of them just use WhatsApp, that's it. If it takes more than a few seconds they won't do it, and then we're back where we started."

2.2 Follow-up email from Mr Tariq, two days later

> **Subject:** RE: equipment thing
>
> Hi,
>
> Forgot to mention a few things.
>
> The drone is different, that needs Hamza's approval before anyone takes it, he is the only licensed operator. Same for the new rover maybe, I will decide later.
>
> We have 4 tripods and 6 field tablets, those are not individually tracked, we just need to know how many are left.
>
> Also sometimes equipment comes back damaged and nobody says anything until the next person takes it to site and finds out. That has cost us two days of client work this year.
>
> It would be good if the system reminds people. We mostly use WhatsApp for everything here.
>
> And I want to be able to look back at last month and see what was out and who had it.
>
> Thanks Tariq

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3. What you have to build

A working application that addresses Horizon's problem, built with:

- **Backend:** .NET 8 or .NET 9, C#, ASP.NET Core Web API
- **Data:** any relational database — SQLite or SQL Server LocalDB is completely fine, and is the sensible choice for a task this size
- **Frontend:** your choice, including plain HTML and JavaScript. It must be usable, but we are not assessing visual design. A beautiful UI over a confused data model scores worse than a plain UI over a sound one.
- **Running:** one documented command, or a short, accurate list of steps in the README. If we cannot run it in five minutes, we review it as if it does not run.

Scope guidance. Not everything mentioned above has to exist in working code. Deciding what to build first, and being able to defend that order, is part of what we are assessing. What you deliberately left out — and why — is as interesting to us as what you built.

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4. What you must submit

A single repository (private GitHub repo with access for us, or a zip file) laid out like this:

```
/src                your application code
/docs               your BMAD artifacts – brief, PRD, epics, stories, architecture
/conversations      every AI conversation, exported in full
/DECISIONS.md       your assumptions and decisions log  (see below)
/README.md          how to run it; what is done; what is not done; what you cut and
                    why; roughly how many hours you spent; which AI tools you used
```

4.1 `DECISIONS.md` — the most important file in your submission

For every point where the requirements did not tell you what to do, record:

Field	What to write
The gap	What was unclear, missing or contradictory — quote the line
What I decided	The decision you made
The alternative	The other reasonable option you considered
Why this one	Your reasoning
What would change my mind	What would have to be true for the other option to win

We expect somewhere between 8 and 20 of these. If you have two, you did not read carefully enough. If you have sixty, you did not prioritise.

4.2 `/conversations` — all of it

Export every AI conversation you used for any part of this task: requirements, BMAD, design, code, debugging, documentation. Markdown, text or JSON — any readable format.

Do not clean them up. Do not delete the parts where it went wrong. The conversation where you spent an hour going in circles is more useful to us than the one that worked first time.

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5. Rules

1. **Do not contact us with questions about the requirements.** Email HR only about logistics — deadline, format, access problems.
2. Use any AI tools you like. Tell us in the README which ones you used and for what.
3. The work must be yours in the sense that you directed it and you understand it. In the interview we will ask you to explain any part of the code and to change it live. If you cannot navigate your own submission, the submission does not count.
4. Tell us in the README roughly how many hours you spent. We are not checking it against a target — there isn't one. We just want your own account of it alongside the work.

Good luck.